

Subject Description Form

Subject Code	ITC3069T
Subject Title	Quality Evaluation for Fashion and Textiles
Credit Value	3
Level	3
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: ITC2203T Foundations of Textiles II Exclusion: SFT308FI Fashion Quality Evaluation and Management
Objectives	The subject gives operational and analytical insights into the areas of fashion product quality improvement within the textile and clothing companies. It provides the essential technical skills to evaluate and manage the performance quality of textile products.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: (a) know the models, dimensions, and management tools of quality (b) identify, select and access (source) the most appropriate international standard tests and test methods to assess the serviceability of a specific fashion product and end-use; (c) conduct and apply the standard quality control tests appropriate to fashion products and interpret and evaluate the results in terms of the relevant national and international performance requirements, standards and specifications for different fashion products; (d) apply critical and creative thinking for identifying, analysing and solving quality and general technical problems related to fashion products; (e) competently and professionally present quality assessment information and results and solutions to quality related problems and ways of improving quality management.
Subject Synopsis/ Indicative Syllabus	(I) Physical Testing of Fashion and Textile Products <i>Textile Properties that Influence Quality Performance:</i> Physical testing of fibre and yarn, such as strength, twist and fineness. Physical testing of fabric such as tensile, tearing and bursting. <i>Abrasion and Wear Properties Evaluation:</i> Abrasion resistance testing. Appearance and colour change due to abrasion, e.g. snagging and pilling. <i>Dimensional Stability Properties Evaluation:</i> Testing of shape and dimensional change in laundering and dry cleaning. Care labelling.

	<i>Fabric hand and comfort evaluation:</i> Thermal conductivity, air and moisture permeability, FAST and KES-F systems. (II) Chemical Testing of Fashion and Textile Products <i>Identification of Fibre Content</i> Systematic identification of fibre types for fashion and textile products, and quantitative analysis of fibre blend content using physical and chemical techniques. <i>Color Difference Evaluation with Instruments</i> Evaluation of color uniformness of fashion and textile products via subjective and objective ways, using grey scales, standard viewing cabinet, and spectrophotometer. <i>Standard Tests for Colour Fastness</i> Importance and principles of evaluating the colour fastness of dyed textiles to light, washing, perspiration and rubbing, and explaining the results. <i>Analysis of Dyestuffs</i> Systematic identification of dyes by chemical methods according to application classes. Evaluation of dyeing and printing faults and raising alternative improvements. (III) Quality Management in Fashion and Textiles <i>Introduction to Quality Management</i> Definition of quality. Importance of quality management. Dimensions of quality. <i>Quality Management Tools</i> Quality improvement methodologies. Quality standard and performance requirement for apparels and other textile products. Total quality management.
Teaching/Learning Methodology	Teaching will be conducted in the form of lectures and laboratory sessions. Lectures will be used primarily to cover a large part of the subject matter. Some appropriate topics with guiding questions will be assigned in advance to students for discussion in lecture classes. Students may also present verbal reports of their essay assignments followed by discussion in class. Demonstrations of physical and chemical testing will be conducted in laboratories to supplement the lecture materials, so as to enhance the student understanding and practice skills on standard testing operations as well as quality management, and to cultivate their problem-solving ability.

Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)				
			a	b	c	d	e
	Continuous Assessment	50%	✓	✓	✓	✓	✓
	1. Test	25%	✓	✓	✓	✓	✓
	2. Project/Report	25%	✓	✓	✓	✓	✓
	Examination	50%	✓	✓	✓	✓	✓
	Total	100%					
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: This subject will provide the students with fundamental theory and basic laboratory briefing so that the students will easily understand the physical testing and chemical testing of fashion and textile products. Coursework such as laboratory reports, group projects and lecture quiz will be used to assess the student’s learning outcomes at different stages and aspects of the subject. The examination will be used to assess students’ intellectual competence in recognising problems, analysing and solving problems as well as understanding the overall theory of the subject.						
Student Study Effort Expected	Class contact:						
	• Lecture					24 Hrs.	
	• Laboratory					15 Hrs.	
	Other student study effort:						
	• Assignment and self-study					66 Hrs.	
	Total student study effort					105 Hrs.	

Reading List and References

AATCC (2010), Technical Manual of American Association of Textile Chemists and Colorists. USA.

ASTM (2010), Annual Book of ASTM Standards, Vol. 07.01 Textiles (I) and Vol. 07.02 Textiles (II).

Collier, B. J. and Epps, H. H. (1999), Textile Testing and Analysis, Prentice-Hall Inc, Upper Saddle River, N.J.

Collier, B. J., Bide, M. J. and Tortora, P. G. (2009), Understanding Textiles. 7th Ed., Pearson; Prentice Hall, Upper Saddle River, N.J.

Fan, Q. G. (2005), Chemical Testing of Textiles. Woodhead Publishing Ltd, Cambridge.

Greaves, P. (1996), Identification of Textile Materials. 8th Ed., Textile Institute, London.

Hearle, J. W. S., Lomas, B. and Cooke, W. D. (1998), Atlas of Fibre Fracture and Damage to Textiles. 2nd Ed., CRC Press, US; Woodhead Publishing Ltd, Cambridge.

Saville, B. P. (1999), Physical Testing of Textiles. Woodhead Publishing, Ltd, in association with the Textile Institute, CRC Press, North and South America.

Kadolph, S. J. (2007), Quality Assurance for Textiles and Apparel. 2nd Edition, Fairchild Publications, NY.

Evans, J. R. (2017), Managing for quality and performance excellence. 10th Edition, Cengage Learning, Boston, MA.